MECHANICS APPLIED TO AEROSPACE ENGINEERING — PART I

# Topic 2: Point Particle Dynamics

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Ground Truth: slides/02.PointParticleDynamics.pdf & teoria/Notes.pdf (Chapter 4)

## 1. Mass, Linear Momentum, and Kinetic Energy

While kinematics describes the geometry of trajectories without regard to the physical influences causing motion, **dynamics** investigates the fundamental physical causes of motion: the forces interacting between matter.

In Newtonian dynamics, every material point $P$ is endowed with an invariant intrinsic scalar property: its **inertial mass** $m_P$ $[\mathrm{kg}]$, which quantitatively measures its resistance to changes in its state of motion.

CORE DYNAMICAL QUANTITIES IN AN INERTIAL REFERENCE FRAME $S_0$

Observed from an inertial reference frame $S_0 = \{O; \mathcal{B}_0\}$, a particle of mass $m_P$ moving at velocity $\mathbf{v}_0^P$ possesses:

- **Linear Momentum (Quantity of Motion) $\mathbf{p}_0$ (Slide 3):**

$$\mathbf{p}_0 = m_P \, \mathbf{v}_0^P \quad [\mathrm{kg \cdot m/s}]$$

  A vector quantity strictly parallel to the instantaneous velocity vector.

- **Kinetic Energy $T_0$ (Slide 3):**

$$T_0 = \frac{1}{2} m_P \, \|\mathbf{v}_0^P\|^2 = \frac{1}{2} m_P \, v_0^2 \geq 0 \quad [\mathrm{J = kg \cdot m^2/s^2}]$$

  A non-negative scalar quantity proportional to the square of the particle's speed $v_0$.

REFERENCE FRAME DEPENDENCE

Unlike mass $m_P$, both $\mathbf{p}_0$ and $T_0$ depend directly on the chosen observer's reference frame $S_0$. If an observer travels alongside the particle, its relative velocity is zero, and thus its measured momentum and kinetic energy vanish.

## 2. Newton's Three Laws of Motion

Classical dynamics is structured upon three empirical laws formulated by Sir Isaac Newton in his *Philosophiae Naturalis Principia Mathematica* (1687):

### 2.1 First Law (Principle of Inertia — Slide 4)

*"A particle continues in its state of rest or of uniform rectilinear motion unless acted upon by a non-zero net external force."*

$$\sum \mathbf{F} = \mathbf{0} \implies \mathbf{p}_0 = \text{constant} \iff \mathbf{v}_0^P = \text{constant} \iff \mathbf{a}_0^P = \mathbf{0}$$

*Mathematical formulation of the First Law in an inertial reference frame*

**Axiomatic Importance:** The First Law establishes the very existence of **inertial reference frames**: spatial reference frames in which unforced bodies do not accelerate.

### 2.2 Second Law (Fundamental Law of Dynamics — Slide 5)

*"The time rate of change of linear momentum of a particle measured in an inertial frame is directly proportional to, and in the direction of, the resultant force acting upon it."*

$$\mathbf{F} = \left.\frac{d\mathbf{p}_0}{dt}\right|_0 = \left.\frac{d(m_P \mathbf{v}_0^P)}{dt}\right|_0$$

*Newton's Second Law for variable or constant mass*

For a classical point particle whose mass $m_P$ remains strictly constant in time ($\dot{m}_P = 0$):

$$\mathbf{F} = m_P \left.\frac{d\mathbf{v}_0^P}{dt}\right|_0 = m_P \, \mathbf{a}_0^P$$

where $\mathbf{F} = \sum_i \mathbf{F}_i$ is the vector sum of all concurrent forces acting on $P$.

### 2.3 Third Law (Action-Reaction Principle — Slide 6)

*"To every action there is always an equal and opposite reaction: the mutual interactions between two particles are always equal in magnitude, opposite in direction, and collinear along their line of centers."*

$$\mathbf{F}_{PQ} = -\mathbf{F}_{QP}, \quad \text{collinear with } (\mathbf{r}^Q - \mathbf{r}^P)$$

*Action-reaction force pair between particles P and Q*

## 3. Systems of Particles and Momentum Conservation

Consider a system composed of $N$ point particles $\{P_1, P_2, \dots, P_N\}$. The net force on particle $i$ decomposes into external forces $\mathbf{F}_i^{\text{ext}}$ exerted by the environment and internal interaction forces $\mathbf{F}_{ij}^{\text{int}}$ exerted by other particles within the system (Slide 7):

$$\mathbf{F}_i = \mathbf{F}_i^{\text{ext}} + \sum_{j \neq i}^{N} \mathbf{F}_{ij}^{\text{int}}$$

### Total Linear Momentum and Conservation (Slide 8)

The total linear momentum of the particle ensemble is the vector sum:

$$\mathbf{p}_{\text{tot}} = \sum_{i=1}^{N} \mathbf{p}_i = \sum_{i=1}^{N} m_i \, \mathbf{v}_0^{P_i}$$

Differentiating with respect to time in the inertial frame $S_0$:

$$\left. \frac{d\mathbf{p}_{\text{tot}}}{dt} \right|_0 = \sum_{i=1}^{N} \mathbf{F}_i = \sum_{i=1}^{N} \mathbf{F}_i^{\text{ext}} + \sum_{i=1}^{N} \sum_{j \neq i}^{N} \mathbf{F}_{ij}^{\text{int}}$$

By Newton's Third Law, internal action-reaction pairs cancel pairwise: $\mathbf{F}_{ij}^{\text{int}} + \mathbf{F}_{ji}^{\text{int}} = \mathbf{0}$. Therefore:

$$\left. \frac{d\mathbf{p}_{\text{tot}}}{dt} \right|_0 = \sum_{i=1}^{N} \mathbf{F}_i^{\text{ext}} = \mathbf{F}^{\text{ext}}$$

PRINCIPLE OF CONSERVATION OF TOTAL LINEAR MOMENTUM

If the resultant of all external forces acting on a system vanishes ($\mathbf{F}^{\text{ext}} = \mathbf{0}$, an **isolated system**), the total linear momentum is strictly constant in time:

$$\mathbf{p}_{\text{tot}}(t) = \text{constant vector}$$

For two interacting particles $P$ and $Q$ in isolation: $\mathbf{p}_0^P(t) + \mathbf{p}_0^Q(t) = \text{constant}$.

# 4. Equations of Motion and Trajectory Integration

Newton's second law $\mathbf{F} = m_P \, \mathbf{a}_0^P$ is a second-order vector differential equation. To integrate it analytically or numerically, we project it onto coordinate bases (Slide 9).

## 4.1 Scalar Equations of Motion in Cartesian Coordinates

Projecting onto the canonical Cartesian basis $\{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$:

$$\begin{cases} m_P \, \ddot{x}(t) = F_x(t, x, y, z, \dot{x}, \dot{y}, \dot{z}) \\ m_P \, \ddot{y}(t) = F_y(t, x, y, z, \dot{x}, \dot{y}, \dot{z}) \\ m_P \, \ddot{z}(t) = F_z(t, x, y, z, \dot{x}, \dot{y}, \dot{z}) \end{cases}$$

*System of three coupled second-order scalar ODEs governing particle dynamics*

## 4.2 State-Space Form and Initial Conditions (Slide 10)

The 3 second-order ODEs can be recast as a system of **6 first-order ordinary differential equations**:

$$\begin{cases} \dot{x} = v_x, & \dot{v}_x = \frac{1}{m_P} F_x(t, x, y, z, v_x, v_y, v_z) \\ \dot{y} = v_y, & \dot{v}_y = \frac{1}{m_P} F_y(t, x, y, z, v_x, v_y, v_z) \\ \dot{z} = v_z, & \dot{v}_z = \frac{1}{m_P} F_z(t, x, y, z, v_x, v_y, v_z) \end{cases}$$

*Standard state-space formulation suitable for integration and numerical solvers*

The general solution contains **6 integration constants**, which are uniquely determined by the **initial conditions** at $t = 0$:

$$\mathbf{r}_0^P(0) = [x_0, y_0, z_0]^T, \quad \mathbf{v}_0^P(0) = [\dot{x}_0, \dot{y}_0, \dot{z}_0]^T$$

## 4.3 Official Slide Benchmark Examples (Explicit Calculus & Chain Rule)

**Example 1: Trajectory Under Gravity (Slide 11)**

Slide 11

Direct time-integration via the Fundamental Theorem of Calculus (Barrow's Rule)

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**Problem Statement:** A point particle $P$ ($m_P = 1$ kg) moves under uniform downward gravity $\mathbf{F} = -m_P g \, \mathbf{j}_0$ ($g = 10$ m/s$^2$). At $t = 0$, the particle is launched from the origin $\mathbf{r}_0^P(0) = \mathbf{0}$ with initial velocity $\mathbf{v}_0^P(0) = 2\,\mathbf{i}_0 + 5\,\mathbf{j}_0$ m/s. Determine the velocity field $\mathbf{v}_0^P(t)$, the position trajectory $\mathbf{r}_0^P(t)$, and the explicit path equation $y(x)$.

**Step 1: Horizontal Motion Along $\mathbf{i}_0$ ($x$-axis)**

Applying Newton's second law along the $x$-direction where no horizontal force acts ($F_x = 0$):

$$m_P\,\ddot{x}(t) = 0 \implies \ddot{x}(t) = \frac{d\dot{x}}{dt} = 0$$

We integrate the acceleration over time from $t' = 0$ to $t' = t$ applying the Fundamental Theorem of Calculus:

$$\int_0^t \frac{d\dot{x}}{dt'}\,dt' = \int_0^t 0\,dt' \implies \Big[\dot{x}(t')\Big]_0^t = 0 \implies \dot{x}(t) - \dot{x}(0) = 0$$

Substituting the initial horizontal velocity $\dot{x}(0) = 2\ \mathrm{m/s}$:

$$\dot{x}(t) = \dot{x}(0) = 2\ \mathrm{m/s} \quad \text{(uniform rectilinear motion)}$$

Next, because $\dot{x}(t) = \frac{dx}{dt}$, we integrate velocity over time:

$$\int_0^t \frac{dx}{dt'}\,dt' = \int_0^t 2\,dt' \implies \Big[x(t')\Big]_0^t = \Big[2t'\Big]_0^t \implies x(t) - x(0) = 2t - 0$$

With $x(0) = 0$, we obtain the parametric horizontal position:

$$x(t) = 2t \quad [\mathrm{m}]$$

**Step 2: Vertical Motion Along $\mathbf{j}_0$ ($y$-axis)**

Projecting Newton's second law onto the vertical axis where weight acts downwards ($F_y = -m_P g$):

$$m_P\,\ddot{y}(t) = -m_P g \implies \ddot{y}(t) = \frac{d\dot{y}}{dt} = -g = -10\ \mathrm{m/s^2}$$

Integrating acceleration over time from $t' = 0$ to $t' = t$:

$$\int_0^t \frac{d\dot{y}}{dt'}\,dt' = \int_0^t (-10)\,dt' \implies \Big[\dot{y}(t')\Big]_0^t = \Big[-10t'\Big]_0^t \implies \dot{y}(t) - \dot{y}(0) = -10t$$

Substituting the initial vertical velocity $\dot{y}(0) = 5\ \mathrm{m/s}$:

$$\dot{y}(t) = 5 - 10t \quad [\mathrm{m/s}]$$

Because $\dot{y}(t) = \frac{dy}{dt}$, we integrate once more with respect to time applying the power rule of integration ( $\int t'dt' = \frac{(t')^2}{2}$):

$$\int_0^t \frac{dy}{dt'}\,dt' = \int_0^t (5 - 10t')\,dt' \implies \Big[y(t')\Big]_0^t = \left[5t' - 10\frac{(t')^2}{2}\right]_0^t$$

$$y(t) - y(0) = \left(5t - 5t^2\right) - (0 - 0)$$

With $y(0) = 0$, we find the parametric vertical position:

$$y(t) = 5t - 5t^2 \quad [\mathrm{m}]$$

**Step 3: Trajectory Cartesian Curve $y(x)$**
To find the geometric path independent of time, we eliminate parameter $t$:

$$x(t) = 2t \implies t(x) = \frac{x}{2}$$

Substituting $t(x)$ directly into $y(t)$:

$$y(x) = 5\left(\frac{x}{2}\right) - 5\left(\frac{x}{2}\right)^2 = \frac{5}{2}x - 5\left(\frac{x^2}{4}\right) = \frac{5}{2}x - \frac{5}{4}x^2$$

This equation represents a classical concave downward parabola with vertex at $x_v = 1\ \mathrm{m}, y_v = 1.25\ \mathrm{m}$.

**Example 2: Inverse Dynamics (Slide 12)**

Slide 12

Finding the resultant force from prescribed kinematics via explicit Chain Rule

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**Problem Statement:** What instantaneous resultant force $\mathbf{F}(t)$ must act on a particle of mass $m_P = 2$ kg so that its velocity vector evolves strictly as $\mathbf{v}_0^P(t) = 2at^2\,\mathbf{i}_0 + e^{bt}\,\mathbf{j}_0$ (where $a, b$ are positive dimensional constants)?

**Pedagogical Rationale:** In inverse dynamics, the kinematics is known and we apply Newton's second law $\mathbf{F} = m_P\,\mathbf{a}_0^P = m_P\,\frac{d\mathbf{v}_0^P}{dt}$ to deduce the required force field. Because the Cartesian unit vectors $\{\mathbf{i}_0, \mathbf{j}_0\}$ are stationary in inertial frame $S_0$ ($\frac{d\mathbf{i}_0}{dt} = \mathbf{0}$, $\frac{d\mathbf{j}_0}{dt} = \mathbf{0}$), we differentiate each scalar component individually.

**Step 1: Differentiation of the $x$-component (Power Rule)**

$$\frac{d}{dt}\left(2at^2\,\mathbf{i}_0\right) = 2a\left(\frac{d(t^2)}{dt}\right)\mathbf{i}_0 = 2a(2t)\,\mathbf{i}_0 = 4at\,\mathbf{i}_0$$

**Step 2: Differentiation of the $y$-component (Chain Rule)**

The vertical velocity component involves a composite exponential function $f(t) = e^{bt}$. We define the intermediate variable $u(t) = bt$. Applying the **chain rule of calculus** $\frac{df}{dt} = \frac{df}{du}\frac{du}{dt}$:

$$u(t) = bt \implies \frac{du}{dt} = \frac{d}{dt}(bt) = b$$

$$\frac{d}{dt}\left(e^{bt}\right) = \frac{d(e^u)}{du}\frac{du}{dt} = e^u \cdot b = b\,e^{bt}$$

Therefore, the vertical acceleration vector component is:

$$\frac{d}{dt}\left(e^{bt}\,\mathbf{j}_0\right) = b\,e^{bt}\,\mathbf{j}_0$$

**Step 3: Synthesis of Resultant Force Vector**

Combining both components, the acceleration of particle $P$ is:

$$\mathbf{a}_0^P(t) = \frac{d\mathbf{v}_0^P}{dt} = 4at\,\mathbf{i}_0 + b\,e^{bt}\,\mathbf{j}_0$$

Multiplying by the mass of the particle $m_P = 2$ kg:

$$\mathbf{F}(t) = m_P\,\mathbf{a}_0^P(t) = 2\,\left(4at\,\mathbf{i}_0 + b\,e^{bt}\,\mathbf{j}_0\right) = 8at\,\mathbf{i}_0 + 2b\,e^{bt}\,\mathbf{j}_0 \quad [\text{N}]$$

# 5. Power, Work, and the Kinetic Energy Theorem

## 5.1 Instantaneous Power (Slide 13)

The power $P$ exerted by a force $\mathbf{F}$ on particle $P$ moving with velocity $\mathbf{v}_0^P$ is the scalar product:

$$P = \mathbf{F} \cdot \mathbf{v}_0^P \quad [\mathrm{W} = \mathrm{J/s}]$$

## 5.2 The Kinetic Energy Theorem (Work-Energy Principle)

Taking the time derivative of kinetic energy $T_0 = \frac{1}{2} m_P (\mathbf{v}_0^P \cdot \mathbf{v}_0^P)$:

$$\frac{dT_0}{dt} = m_P \, \mathbf{v}_0^P \cdot \left. \frac{d\mathbf{v}_0^P}{dt} \right|_0 = m_P \, \mathbf{a}_0^P \cdot \mathbf{v}_0^P$$

Substituting Newton's second law $\mathbf{F} = m_P \mathbf{a}_0^P$:

$$\frac{dT_0}{dt} = \mathbf{F} \cdot \mathbf{v}_0^P = P$$

*The time rate of change of kinetic energy equals the instantaneous power exerted by the net force*

## 5.3 Work Done by a Force

Integrating power over the time interval $[t_1, t_2]$:

$$W_{12} = \int_{t_1}^{t_2} P \, dt = \int_{t_1}^{t_2} (\mathbf{F} \cdot \mathbf{v}_0^P) \, dt = \int_{\mathbf{r}_1}^{\mathbf{r}_2} \mathbf{F} \cdot d\mathbf{r} = \Delta T_0 = T_{0,2} - T_{0,1}$$

FUNDAMENTAL TANGENTIAL WORK CRITERION

Expressing velocity in the intrinsic Frenet-Serret frame ($\mathbf{v}_0^P = v_0 \, \mathbf{e}_t$), power becomes:

$$P = \mathbf{F} \cdot (v_0 \, \mathbf{e}_t) = F_t \, v_0$$

**A force exerts non-zero power and performs work if and only if it produces tangential acceleration (** $F_t \neq 0$**).** Forces perpendicular to the trajectory (normal constraints, string tension, magnetic Lorentz forces) do **identically zero work** ($W \equiv 0$).

## 6. Conservative Forces and Potential Energy

In general, the work integral $W_{12} = \int_\Gamma \mathbf{F} \cdot d\mathbf{r}$ depends on the path $\Gamma$ connecting the endpoints.

### 6.1 Conservative Forces (Slide 14)

A force $\mathbf{F}$ is **conservative** if the work it performs between two points is **strictly path-independent**:

$$\int_{\Gamma_A} \mathbf{F} \cdot d\mathbf{r} = \int_{\Gamma_B} \mathbf{F} \cdot d\mathbf{r} \iff \oint_{\mathcal{C}} \mathbf{F} \cdot d\mathbf{r} = 0 \quad \text{along any closed curve } \mathcal{C}$$

### 6.2 Potential Energy Function $V_0(\mathbf{r})$

Fixing an arbitrary reference position $\mathbf{r}_*$, the potential energy is defined by:

$$V_0(\mathbf{r}) = -\int_{\mathbf{r}_*}^{\mathbf{r}} \mathbf{F} \cdot d\mathbf{r}'$$

The work done by a conservative force equals the negative change in potential energy:

$$W_{12} = -(V_0(\mathbf{r}_2) - V_0(\mathbf{r}_1)) = -\Delta V_0$$

Differentiating: $dW = \mathbf{F} \cdot d\mathbf{r} = -dV_0 = -\nabla V_0 \cdot d\mathbf{r}$, establishing:

$$\mathbf{F} = -\nabla V_0 = -\begin{bmatrix} \frac{\partial V_0}{\partial x} \\ \frac{\partial V_0}{\partial y} \\ \frac{\partial V_0}{\partial z} \end{bmatrix}$$

*A conservative force is always the negative gradient of its scalar potential energy function*

### 6.3 Mathematical Condition: The Curl Test (Slide 16)

In a simply-connected domain, a continuously differentiable force field $\mathbf{F}(\mathbf{r})$ is conservative **if and only if its curl vanishes identically**:

$$\nabla \times \mathbf{F} = \mathbf{0} \iff \operatorname{curl}(\mathbf{F}) = \mathbf{0}$$

### 6.4 Fundamental Aerospace Potential Functions (Slide 16)

| Physical Force | Vector Expression F | Potential Energy $V_0(\mathbf{r})$ |
|---|---|---|
| **Uniform Gravity** | $\mathbf{W} = -m_P g\,\mathbf{k}_0$ | $V_0(z) = m_P g z + C$ |
| **Linear Spring (zero natural length)** | $\mathbf{F} = -k\,\mathbf{r}_0^P$ | $V_0(\mathbf{r}) = \frac{1}{2}k\,\|\mathbf{r}_0^P\|^2 = \frac{1}{2}k\,r^2$ |
| **Inverse-Square Central Gravity** | $\mathbf{F} = -\frac{GMm_P}{r^2}\mathbf{e}_r$ | $V_0(r) = -\frac{GMm_P}{r}$ |

## 7. Polar Vector Field Curl Tests

In cylindrical / polar coordinates $\{R, \theta, z\}$, the curl of a vector field is given by:

$$\nabla \times \mathbf{F} = \left(\frac{1}{R}\frac{\partial F_z}{\partial \theta} - \frac{\partial F_\theta}{\partial z}\right)\mathbf{e}_R + \left(\frac{\partial F_R}{\partial z} - \frac{\partial F_z}{\partial R}\right)\mathbf{e}_\theta + \frac{1}{R}\left(\frac{\partial (RF_\theta)}{\partial R} - \frac{\partial F_R}{\partial \theta}\right)\mathbf{e}_z$$

**Example 1: Radial Force $\mathbf{F} = aR\,\mathbf{e}_R$ (Slide 17)**

`Conservative`

Is it conservative? If so, compute $V_0(R)$

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**Evaluation:** Here $F_R = aR$, $F_\theta = 0$, $F_z = 0$.

$$(\nabla \times \mathbf{F})_z = \frac{1}{R}\left(\frac{\partial (R\cdot 0)}{\partial R} - \frac{\partial (aR)}{\partial \theta}\right) = \frac{1}{R}(0 - 0) = 0$$

Since all components of $\nabla \times \mathbf{F}$ vanish identically, $\mathbf{F} = aR\,\mathbf{e}_R$ is **conservative**.

To find $V_0(R)$: $\mathbf{F} = -\nabla V_0 = -\frac{\partial V_0}{\partial R}\mathbf{e}_R \implies \frac{\partial V_0}{\partial R} = -aR$:

$$V_0(R) = -\int aR\,dR = -\frac{1}{2}aR^2 + C$$

**Example 2: Azimuthal Force $\mathbf{F} = bR^2\,\mathbf{e}_\theta$ (Slide 18)**

`Non-Conservative`

Is it conservative?

---

**Evaluation:** Here $F_R = 0$, $F_\theta = bR^2$, $F_z = 0$.

$$(\nabla \times \mathbf{F})_z = \frac{1}{R}\left(\frac{\partial (R \cdot bR^2)}{\partial R} - \frac{\partial (0)}{\partial \theta}\right) = \frac{1}{R}\frac{\partial (bR^3)}{\partial R} = \frac{1}{R}(3bR^2) = 3bR \neq 0$$

Because the curl is non-zero ($\nabla \times \mathbf{F} = 3bR\,\mathbf{e}_z \neq \mathbf{0}$), the force is **non-conservative** and no potential energy function can exist.

## 8. Mechanical Energy and 1D Energy Diagrams

Let a particle be acted upon simultaneously by conservative forces $\mathbf{F}_C = -\nabla V_0$ and non-conservative forces $\mathbf{F}_N$ (Slide 19):

$$m_P\,\mathbf{a}_0^P = \mathbf{F}_C + \mathbf{F}_N = -\nabla V_0 + \mathbf{F}_N$$

Taking the scalar product with velocity $\mathbf{v}_0^P$:

$$\frac{dT_0}{dt} = -\nabla V_0 \cdot \mathbf{v}_0^P + \mathbf{F}_N \cdot \mathbf{v}_0^P = -\frac{dV_0}{dt} + \mathbf{F}_N \cdot \mathbf{v}_0^P$$

Defining the **Total Mechanical Energy** $E_0$:

$$E_0 = T_0 + V_0 \implies \frac{dE_0}{dt} = \mathbf{F}_N \cdot \mathbf{v}_0^P$$

*The rate of change of mechanical energy equals the power exerted by non-conservative forces*

If non-conservative forces do zero work ($\mathbf{F}_N \cdot \mathbf{v}_0^P = 0$), then $E_0 = \text{constant}$ (a **first integral of motion**).

### 8.1 1D Qualitative Motion Analysis (Slide 20)

In a 1D conservative system:

$$T_0(x) = \frac{1}{2} m_P\, v^2 = E_0 - V_0(x) \geq 0$$

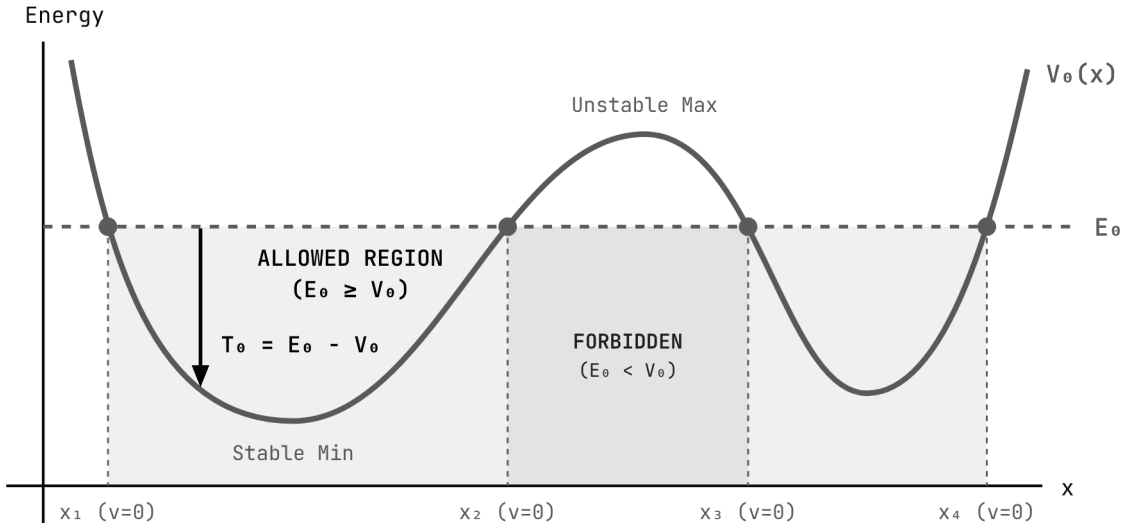



*Figure 2.1: 1D energy diagram illustrating total energy $E_0$, potential energy $V_0(x)$, allowed regions, forbidden regions, and turning points.*

- **Allowed Regions ($E_0 \geq V_0$):** Domains where physical motion can take place.
- **Forbidden Regions ($E_0 < V_0$):** Domains where kinetic energy would be negative ($v^2 < 0$), physically inaccessible.
- **Turning Points ($x_{\text{turn}}$):** Points where $E_0 = V_0(x_{\text{turn}}) \implies T_0 = 0 \implies v = 0$. The particle stops and reverses its direction of motion.
- **Instantaneous Speed:** $v(x) = \sqrt{\frac{2(E_0 - V_0(x))}{m_P}}$. Speed is maximal at the local minimum of $V_0(x)$.

## 8.2 Slide 15 Potential Example (Explicit Chain Rule Derivation)

**Force Field from 1D Potential Function (Slide 15)**

Slide 15

Analytical differentiation using the chain rule

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**Problem:** Determine the 1D force $F(x)$ associated with the potential energy function:

$$V_0(x) = \cos(x) + \cos\left(\frac{x}{2} - \frac{\pi}{4}\right)$$

**Pedagogical Rationale:** In one dimension, a conservative force is defined by the negative spatial derivative of potential energy: $F(x) = -\frac{dV_0}{dx}$. We differentiate each term separately:

1. **First Term:** The elementary derivative of cosine is:

$$\frac{d}{dx}\big[\cos(x)\big] = -\sin(x)$$

2. **Second Term (Chain Rule):** The argument is an affine function of $x$. We set the intermediate variable:

$$u(x) = \frac{x}{2} - \frac{\pi}{4} \implies \frac{du}{dx} = \frac{d}{dx}\left(\frac{x}{2}\right) - \frac{d}{dx}\left(\frac{\pi}{4}\right) = \frac{1}{2} - 0 = \frac{1}{2}$$

Applying the chain rule $\frac{d}{dx}\big[f(u(x))\big] = \frac{df}{du}\frac{du}{dx}$:

$$\frac{d}{dx}\left[\cos\left(\frac{x}{2} - \frac{\pi}{4}\right)\right] = \frac{d(\cos u)}{du}\frac{du}{dx} = (-\sin u)\cdot\left(\frac{1}{2}\right) = -\frac{1}{2}\sin\left(\frac{x}{2} - \frac{\pi}{4}\right)$$

3. **Resultant Force:** Summing both derivatives and applying the fundamental leading minus sign ($F = -dV_0/dx$):

$$F(x) = -\left[-\sin(x) - \frac{1}{2}\sin\left(\frac{x}{2} - \frac{\pi}{4}\right)\right] = \sin(x) + \frac{1}{2}\sin\left(\frac{x}{2} - \frac{\pi}{4}\right)$$

*Physical Interpretation:* The force always acts in the direction of negative potential gradient (directing the particle "downhill" into potential valleys), with magnitude equal to the slope of the curve.

# 9. Equilibrium Configurations and Formal Stability Criteria

## 9.1 Equilibrium Configuration (Slide 21)

A configuration $\mathbf{r}_*$ is an **equilibrium configuration** if a particle released at $\mathbf{r}_*$ with zero velocity remains permanently at rest:

$$\mathbf{a}_0^P = \mathbf{0} \iff \sum \mathbf{F}(\mathbf{r}_*) = \mathbf{0} \iff \nabla V_0(\mathbf{r}_*) = \mathbf{0}$$

In 1D, equilibrium positions correspond to the critical points of the potential energy function: $\frac{dV_0}{dx}\big|_{x_*} = 0$.

### 9.2 Formal Definition of Stability (Lyapunov Criterion)

An equilibrium position $\mathbf{r}_*$ is **stable** if for any arbitrarily chosen position and velocity tolerances $\varepsilon_r, \varepsilon_v > 0$, there exist bounds $\delta_r, \delta_v > 0$ such that if the particle is perturbed initially with $\|\mathbf{r}(0) - \mathbf{r}_*\| < \delta_r$ and $\|\mathbf{v}(0)\| < \delta_v$, then:

$$\|\mathbf{r}(t) - \mathbf{r}_*\| < \varepsilon_r \quad \text{and} \quad \|\mathbf{v}(t)\| < \varepsilon_v \quad \forall t \geq 0$$

Otherwise, the equilibrium configuration is **unstable**.

### 9.3 Second-Derivative Curvature Criterion in 1D (Slide 22)

Expanding $V_0(x)$ in a Taylor series about equilibrium $x_*$:

$$F(x) = -\frac{dV_0}{dx} \approx -\left.\frac{d^2V_0}{dx^2}\right|_{x_*} (x - x_*)$$

- **Stable Equilibrium ($\left.\frac{d^2V_0}{dx^2}\right|_{x_*} > 0$):** $x_*$ is a **local minimum** (potential well). The force is restoring ($F \approx -k(x - x_*)$), pushing the particle back towards $x_*$.
- **Unstable Equilibrium ($\left.\frac{d^2V_0}{dx^2}\right|_{x_*} < 0$):** $x_*$ is a **local maximum** (potential crest). The force is repulsive ($F \approx +k(x - x_*)$), driving the particle away from $x_*$.
- **Neutral / Indeterminate Equilibrium ($\left.\frac{d^2V_0}{dx^2}\right|_{x_*} = 0$):** Inflection point; higher-order derivatives must be inspected.

## 10. Solved Benchmark Problem: The Simple Pendulum (Slide 23 & Notes Sec. 4.7)

ANALYTICAL & PEDAGOGICAL RIGOR AUDIT (PROBLEM_STEP_MENTOR)

This classic problem synthesizes all core concepts of Topic 2: polar kinematic constraints, dynamic force balance, string tension determination, exact mechanical energy first integral, equilibrium stability, and small-amplitude harmonic linearization.

### The Simple Pendulum (Slide 23 & Notes Sec. 4.7)

UC3M Benchmark

Nonlinear dynamics, energy first integral, and harmonic linearization

**Problem Statement:** A point particle $P$ of mass $m$ is suspended from a fixed point $O$ by an inextensible, massless rigid rod of length $\ell$, subject to uniform downward gravity $\mathbf{g} = -g\mathbf{j}_0$. Determine the equations of motion, string tension, conservation of energy, equilibrium stability, and small-amplitude oscillations.

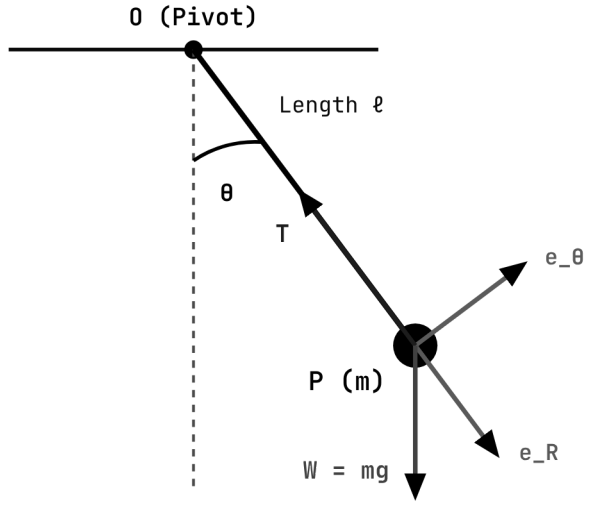



*Figure 2.2: Free body diagram of the simple pendulum with polar unit vectors $\{\mathbf{e}_R, \mathbf{e}_\theta\}$, gravity $\mathbf{W}$, and tension $\mathbf{T}$.*

### Step 1: Coordinates, Basis, and Kinematic Constraints (Explicit Calculus)

**Pedagogical Justification:** The rod maintains a constant distance $R = \ell$ from pivot $O$, constraining the particle to a circular trajectory in the vertical $Oxy$ plane. We define polar coordinates $\{R, \theta\}$ with fixed origin at pivot $O$, equipped with the local orthonormal basis $\{\mathbf{e}_R, \mathbf{e}_\theta\}$. The angle $\theta$ is measured from the downward vertical (so $\theta = 0$ is the hanging equilibrium and $\mathbf{e}_R(0) = -\mathbf{j}_0$), counter-clockwise in the $Oxy$ plane:

$$\mathbf{e}_R(\theta) = \sin\theta\,\mathbf{i}_0 - \cos\theta\,\mathbf{j}_0$$

$$\mathbf{e}_\theta(\theta) = \cos\theta\,\mathbf{i}_0 + \sin\theta\,\mathbf{j}_0$$

We differentiate the rotating basis vectors with respect to time applying the **chain rule** ($\frac{d}{dt} = \frac{d\theta}{dt}\frac{\partial}{\partial\theta} = \dot\theta\frac{\partial}{\partial\theta}$):

$$\frac{d\mathbf{e}_R}{dt} = \frac{\partial\mathbf{e}_R}{\partial\theta}\frac{d\theta}{dt} = \left(\frac{d(\sin\theta)}{d\theta}\mathbf{i}_0 - \frac{d(\cos\theta)}{d\theta}\mathbf{j}_0\right)\dot\theta = (\cos\theta\,\mathbf{i}_0 + \sin\theta\,\mathbf{j}_0)\dot\theta = \dot\theta\,\mathbf{e}_\theta$$

$$\frac{d\mathbf{e}_\theta}{dt} = \frac{\partial\mathbf{e}_\theta}{\partial\theta}\frac{d\theta}{dt} = \left(\frac{d(\cos\theta)}{d\theta}\mathbf{i}_0 + \frac{d(\sin\theta)}{d\theta}\mathbf{j}_0\right)\dot\theta = (-\sin\theta\,\mathbf{i}_0 + \cos\theta\,\mathbf{j}_0)\dot\theta = -\dot\theta\,\mathbf{e}_R$$

The rigidity and inextensibility of the rod impose the holonomic constraint:

$$R(t) = \ell = \text{constant} \implies \dot{R}(t) = 0, \quad \ddot{R}(t) = 0$$

- **Position Vector:**

$$\mathbf{r}_0^P = \ell\, \mathbf{e}_R$$

- **Velocity Vector (Product Rule):**

$$\mathbf{v}_0^P = \frac{d}{dt}\left(\ell\, \mathbf{e}_R\right) = \underbrace{\frac{d\ell}{dt}}_{=0} \mathbf{e}_R + \ell \frac{d\mathbf{e}_R}{dt} = \ell(\dot{\theta}\, \mathbf{e}_\theta) = \ell\dot{\theta}\, \mathbf{e}_\theta$$

- **Acceleration Vector (Product Rule & Chain Rule):**

$$\mathbf{a}_0^P = \frac{d}{dt}\left(\ell\dot{\theta}\, \mathbf{e}_\theta\right) = \ell\left(\frac{d\dot{\theta}}{dt}\mathbf{e}_\theta + \dot{\theta}\frac{d\mathbf{e}_\theta}{dt}\right) = \ell\left(\ddot{\theta}\, \mathbf{e}_\theta + \dot{\theta}(-\dot{\theta}\, \mathbf{e}_R)\right) = -\ell\dot{\theta}^2\, \mathbf{e}_R + \ell\ddot{\theta}\, \mathbf{e}_\theta$$

### Step 2: Force Projection and Newton's Second Law

Two forces act on particle $P$:

1. **Rod Tension $\mathbf{T}$:** Acts radially inwards toward pivot $O$:

$$\mathbf{T} = -T\, \mathbf{e}_R \quad (T \geq 0)$$

2. **Weight $\mathbf{W}$:** Acts vertically downwards ($\mathbf{W} = -mg\mathbf{j}_0$). We project it onto the polar basis $\{\mathbf{e}_R, \mathbf{e}_\theta\}$ via orthogonal scalar projections:

$$\mathbf{W} = (\mathbf{W} \cdot \mathbf{e}_R)\mathbf{e}_R + (\mathbf{W} \cdot \mathbf{e}_\theta)\mathbf{e}_\theta$$

$$\mathbf{W} \cdot \mathbf{e}_R = -mg\mathbf{j}_0 \cdot (\sin\theta\, \mathbf{i}_0 - \cos\theta\, \mathbf{j}_0) = mg\cos\theta$$

$$\mathbf{W} \cdot \mathbf{e}_\theta = -mg\mathbf{j}_0 \cdot (\cos\theta\,\mathbf{i}_0 + \sin\theta\,\mathbf{j}_0) = -mg\sin\theta$$

$$\mathbf{W} = mg\cos\theta\,\mathbf{e}_R - mg\sin\theta\,\mathbf{e}_\theta$$

Applying Newton's second law: $m\mathbf{a}_0^P = \mathbf{W} + \mathbf{T}$:

$$m(-\ell\dot{\theta}^2\,\mathbf{e}_R + \ell\ddot{\theta}\,\mathbf{e}_\theta) = (mg\cos\theta - T)\,\mathbf{e}_R - mg\sin\theta\,\mathbf{e}_\theta$$

Equating components along each basis vector:

- **Radial Component ($\mathbf{e}_R$):**

$$-m\ell\dot{\theta}^2 = mg\cos\theta - T \implies T(\theta,\dot{\theta}) = m(g\cos\theta + \ell\dot{\theta}^2)$$

  *Physical insight:* The rod tension must simultaneously offset the radial gravitational component $mg\cos\theta$ and supply the centripetal acceleration $m\ell\dot{\theta}^2$.

- **Azimuthal Component ($\mathbf{e}_\theta$):**

$$m\ell\ddot{\theta} = -mg\sin\theta \implies \ddot{\theta} + \frac{g}{\ell}\sin\theta = 0$$

  *Physical insight:* This is the exact, nonlinear second-order differential equation of motion of the simple pendulum.

### Step 3: Mechanical Energy Conservation, Explicit Time Derivative, & First Integral

Because tension $\mathbf{T}$ acts perpendicular to the instantaneous velocity ($\mathbf{T}\cdot\mathbf{v}_0^P = (-T\mathbf{e}_R)\cdot(\ell\dot{\theta}\mathbf{e}_\theta) = 0$), non-conservative work vanishes identically. Mechanical energy $E_0$ is strictly conserved:

- **Kinetic Energy:** $T_0 = \frac{1}{2}m\|\mathbf{v}_0^P\|^2 = \frac{1}{2}m(\ell\dot{\theta})^2 = \frac{1}{2}m\ell^2\dot{\theta}^2$
- **Potential Energy:** Setting $V_0 = 0$ at the lowest point ($\theta = 0$): $V_0(\theta) = mg\ell(1-\cos\theta)$
- **Total Mechanical Energy:**

$$E_0 = T_0 + V_0 = \frac{1}{2}m\ell^2\dot{\theta}^2 + mg\ell(1-\cos\theta) = \text{constant}$$

**Rigorous Proof of Conservation via the Chain Rule:** Differentiating $E_0$ with respect to time $t$:

$$\frac{dT_0}{dt} = \frac{d}{dt}\left(\frac{1}{2}m\ell^2\dot{\theta}^2\right) = \frac{1}{2}m\ell^2\left(\frac{d(\dot{\theta}^2)}{d\dot{\theta}}\frac{d\dot{\theta}}{dt}\right) = \frac{1}{2}m\ell^2(2\dot{\theta}\ddot{\theta}) = m\ell^2\dot{\theta}\ddot{\theta}$$

$$\frac{dV_0}{dt} = \frac{d}{dt}\left[mg\ell(1-\cos\theta)\right] = mg\ell\left(\frac{d(1-\cos\theta)}{d\theta}\frac{d\theta}{dt}\right) = mg\ell(\sin\theta)\dot{\theta}$$

Summing both terms:

$$\frac{dE_0}{dt} = m\ell^2\dot{\theta}\ddot{\theta} + mg\ell\sin\theta\dot{\theta} = m\ell\dot{\theta}\left(\ell\ddot{\theta} + g\sin\theta\right)$$

Since $\ell\ddot{\theta} + g\sin\theta = 0$ by the equation of motion, $\frac{dE_0}{dt} \equiv 0$ identically!

Solving $E_0 = \frac{1}{2}m\ell^2\dot{\theta}^2 + mg\ell(1-\cos\theta)$ for angular velocity $\dot{\theta}$ yields the **first integral of motion**:

$$\dot{\theta} = \pm\sqrt{\frac{2}{m\ell^2}\left[E_0 - mg\ell(1-\cos\theta)\right]}$$

Turning points occur when $\dot{\theta} = 0 \implies \cos\theta_{\max} = 1 - \frac{E_0}{mg\ell}$.

### Step 4: Equilibrium Configurations and Curvature Stability

Equilibrium positions correspond to the critical points of potential energy $V_0(\theta)$:

$$\frac{dV_0}{d\theta} = mg\ell\sin\theta = 0 \implies \theta_* = 0, \pm\pi$$

We classify stability by evaluating the second derivative (curvature test):

$$\frac{d^2V_0}{d\theta^2} = \frac{d}{d\theta}(mg\ell\sin\theta) = mg\ell\cos\theta$$

- **At $\theta_* = 0$ (hanging vertically downwards):**

$$\left.\frac{d^2V_0}{d\theta^2}\right|_{\theta=0} = mg\ell\cos(0) = +mg\ell > 0 \implies \textbf{Stable Equilibrium (Local Minimum)}$$

- **At $\theta_* = \pm\pi$ (inverted vertically upwards):**

$$\left.\frac{d^2V_0}{d\theta^2}\right|_{\theta=\pm\pi} = mg\ell\cos(\pm\pi) = -mg\ell < 0 \implies \textbf{Unstable Equilibrium (Local Maximum)}$$

### Step 5: Small-Amplitude Linearization ($\theta \ll 1$)

The Maclaurin series expansion of $\sin\theta$ about the stable equilibrium $\theta = 0$ is:

$$\sin\theta = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!}\theta^{2n+1} = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots = \theta - \frac{\theta^3}{6} + \mathcal{O}(\theta^5)$$

For small angular deflections $|\theta| \ll 1$ rad (e.g., $|\theta| < 10^\circ \approx 0.175$ rad), the cubic term is negligible ($\frac{\theta^3}{6} \approx 0.00089 \ll 0.175$). Retaining only the first-order linear term $\sin\theta \approx \theta$, the nonlinear equation reduces to:

$$\ddot{\theta} + \frac{g}{\ell}\theta = 0 \iff \ddot{\theta} + \omega_0^2\theta = 0$$

*Linearized second-order ODE of the simple harmonic oscillator*

The characteristic polynomial is $r^2 + \omega_0^2 = 0 \implies r = \pm i\omega_0$, yielding the general harmonic oscillation:

$$\theta(t) = \theta_{\max}\cos(\omega_0 t + \phi_0)$$

The natural angular frequency $\omega_0$ and period of oscillation $T_0$ are:

$$\omega_0 = \sqrt{\frac{g}{\ell}} \quad [\text{rad/s}], \quad T_0 = \frac{2\pi}{\omega_0} = 2\pi\sqrt{\frac{\ell}{g}} \quad [\text{s}]$$

Notice that period $T_0$ depends strictly on pendulum length $\ell$ and gravitational acceleration $g$, exhibiting complete independence from particle mass $m$ and amplitude $\theta_{\max}$ (isochronism of small oscillations).